MSG No. 160.68 R_I3m [Type IV, trigonal]

Table 1: Wyckoff site: `6a`, site symmetry: `3m`.

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4,5,6]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{3}\right]$	`[7,8,9,10,11,12]`
3	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{2}{3}\right]$	`[13,14,15,16,17,18]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[19,20,21,22,23,24]`
5	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{5}{6}\right]$	`[25,26,27,28,29,30]`
6	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{6}\right]$	`[31,32,33,34,35,36]`

Table 2: Wyckoff site: `18b`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\ -x,\ z]$	`[1,5]`
2	$[x,\ 2x,\ z]$	`[2,6]`
3	$[-2x,\ -x,\ z]$	`[3,4]`
4	$\left[x+\frac{2}{3},\ \frac{1}{3}-x,\ z+\frac{1}{3}\right]$	`[7,11]`
5	$\left[x+\frac{2}{3},\ 2x+\frac{1}{3},\ z+\frac{1}{3}\right]$	`[8,12]`
6	$\left[\frac{2}{3}-2x,\ \frac{1}{3}-x,\ z+\frac{1}{3}\right]$	`[9,10]`
7	$\left[x+\frac{1}{3},\ \frac{2}{3}-x,\ z+\frac{2}{3}\right]$	`[13,17]`
8	$\left[x+\frac{1}{3},\ 2x+\frac{2}{3},\ z+\frac{2}{3}\right]$	`[14,18]`
9	$\left[\frac{1}{3}-2x,\ \frac{2}{3}-x,\ z+\frac{2}{3}\right]$	`[15,16]`
10	$\left[x,\ -x,\ z+\frac{1}{2}\right]$	`[19,23]`
11	$\left[x,\ 2x,\ z+\frac{1}{2}\right]$	`[20,24]`
12	$\left[-2x,\ -x,\ z+\frac{1}{2}\right]$	`[21,22]`
13	$\left[x+\frac{2}{3},\ \frac{1}{3}-x,\ z+\frac{5}{6}\right]$	`[25,29]`
14	$\left[x+\frac{2}{3},\ 2x+\frac{1}{3},\ z+\frac{5}{6}\right]$	`[26,30]`
15	$\left[\frac{2}{3}-2x,\ \frac{1}{3}-x,\ z+\frac{5}{6}\right]$	`[27,28]`
16	$\left[x+\frac{1}{3},\ \frac{2}{3}-x,\ z+\frac{1}{6}\right]$	`[31,35]`
17	$\left[x+\frac{1}{3},\ 2x+\frac{2}{3},\ z+\frac{1}{6}\right]$	`[32,36]`
18	$\left[\frac{1}{3}-2x,\ \frac{2}{3}-x,\ z+\frac{1}{6}\right]$	`[33,34]`

Table 3: Wyckoff site: `36c`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x-y,\ z]$	`[2]`
3	$[-x+y,\ -x,\ z]$	`[3]`
4	$[-x+y,\ y,\ z]$	`[4]`
5	$[-y,\ -x,\ z]$	`[5]`
6	$[x,\ x-y,\ z]$	`[6]`
7	$\left[x+\frac{2}{3},\ y+\frac{1}{3},\ z+\frac{1}{3}\right]$	`[7]`
8	$\left[\frac{2}{3}-y,\ x-y+\frac{1}{3},\ z+\frac{1}{3}\right]$	`[8]`
9	$\left[-x+y+\frac{2}{3},\ \frac{1}{3}-x,\ z+\frac{1}{3}\right]$	`[9]`
10	$\left[-x+y+\frac{2}{3},\ y+\frac{1}{3},\ z+\frac{1}{3}\right]$	`[10]`

continued ...

Table 3

No.	position	mapping
11	$\left[\frac{2}{3}-y,\ \frac{1}{3}-x,\ z+\frac{1}{3}\right]$	[11]
12	$\left[x+\frac{2}{3},\ x-y+\frac{1}{3},\ z+\frac{1}{3}\right]$	[12]
13	$\left[x+\frac{1}{3},\ y+\frac{2}{3},\ z+\frac{2}{3}\right]$	[13]
14	$\left[\frac{1}{3}-y,\ x-y+\frac{2}{3},\ z+\frac{2}{3}\right]$	[14]
15	$\left[-x+y+\frac{1}{3},\ \frac{2}{3}-x,\ z+\frac{2}{3}\right]$	[15]
16	$\left[-x+y+\frac{1}{3},\ y+\frac{2}{3},\ z+\frac{2}{3}\right]$	[16]
17	$\left[\frac{1}{3}-y,\ \frac{2}{3}-x,\ z+\frac{2}{3}\right]$	[17]
18	$\left[x+\frac{1}{3},\ x-y+\frac{2}{3},\ z+\frac{2}{3}\right]$	[18]
19	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[19]
20	$\left[-y,\ x-y,\ z+\frac{1}{2}\right]$	[20]
21	$\left[-x+y,\ -x,\ z+\frac{1}{2}\right]$	[21]
22	$\left[-x+y,\ y,\ z+\frac{1}{2}\right]$	[22]
23	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[23]
24	$\left[x,\ x-y,\ z+\frac{1}{2}\right]$	[24]
25	$\left[x+\frac{2}{3},\ y+\frac{1}{3},\ z+\frac{5}{6}\right]$	[25]
26	$\left[\frac{2}{3}-y,\ x-y+\frac{1}{3},\ z+\frac{5}{6}\right]$	[26]
27	$\left[-x+y+\frac{2}{3},\ \frac{1}{3}-x,\ z+\frac{5}{6}\right]$	[27]
28	$\left[-x+y+\frac{2}{3},\ y+\frac{1}{3},\ z+\frac{5}{6}\right]$	[28]
29	$\left[\frac{2}{3}-y,\ \frac{1}{3}-x,\ z+\frac{5}{6}\right]$	[29]
30	$\left[x+\frac{2}{3},\ x-y+\frac{1}{3},\ z+\frac{5}{6}\right]$	[30]
31	$\left[x+\frac{1}{3},\ y+\frac{2}{3},\ z+\frac{1}{6}\right]$	[31]
32	$\left[\frac{1}{3}-y,\ x-y+\frac{2}{3},\ z+\frac{1}{6}\right]$	[32]
33	$\left[-x+y+\frac{1}{3},\ \frac{2}{3}-x,\ z+\frac{1}{6}\right]$	[33]
34	$\left[-x+y+\frac{1}{3},\ y+\frac{2}{3},\ z+\frac{1}{6}\right]$	[34]
35	$\left[\frac{1}{3}-y,\ \frac{2}{3}-x,\ z+\frac{1}{6}\right]$	[35]
36	$\left[x+\frac{1}{3},\ x-y+\frac{2}{3},\ z+\frac{1}{6}\right]$	[36]